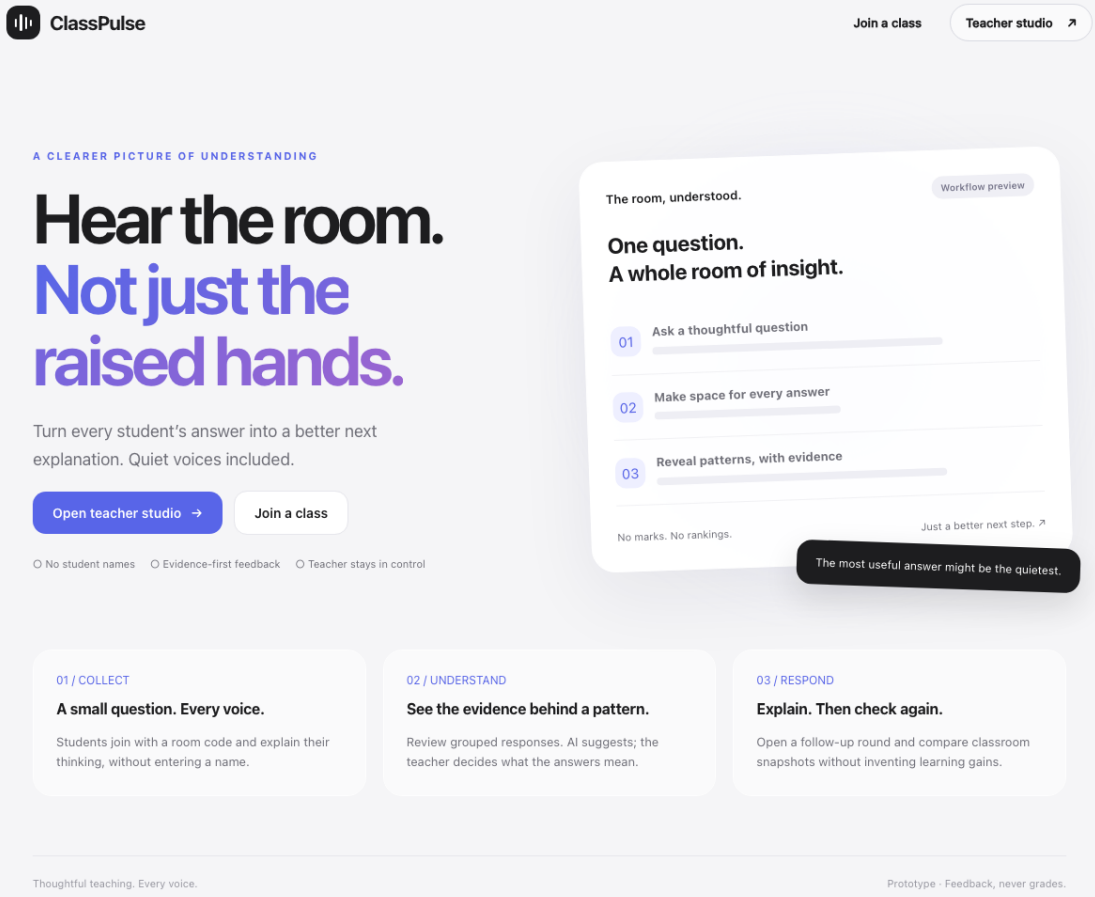
Your first working classroom.

A small question. Every voice. A teacher who can see the evidence.



What you are building: a classroom feedback loop. A teacher asks a question; students answer without entering names; machine learning groups similar wording; the teacher checks the originals and asks again.

What is already built: the app, separate teacher/student views, local ML, evidence review, follow-up rounds, export and a labelled sample class.

Start with this guide. It replaces the Streamlit setup instructions in the earlier planning PDF. Keep README.md for detailed technical instructions.

Install. Then open.

Use a laptop with Python, a browser and internet for the first installation.

1. Get Python. Open python.org/downloads. Install Python 3.13 or 3.14. On Windows, enable Add Python to PATH if offered. Finish the installer.

2. Extract the ZIP. On Windows, right-click and choose Extract All. On Mac, double-click the ZIP. Open the ClassPulse folder. Do not move app.py away from static or templates.

3. Start. Windows: double-click START_WINDOWS.bat. Mac: open START_MAC.command. If needed, open Terminal in this folder and type **zsh START_MAC.command**, then Enter.

4. Wait. The first start downloads dependencies. Keep the terminal open. If an error appears, read the last line before retrying.

5. Choose your teacher password. Use at least 10 characters. Typing is hidden on purpose. Enter it, press Enter, and repeat it. Keep it private.

6. Visit http://localhost:8765. Click Teacher studio and sign in. This address works on this laptop; it is not a public submission link.

If you forget the password: stop the server with Ctrl+C. Mac: run .venv/bin/python setup.py. Windows: run .venv\Scripts\python setup.py. Restart the launcher.

Try both sides.

Use a normal browser window for the teacher and an incognito window for a student.

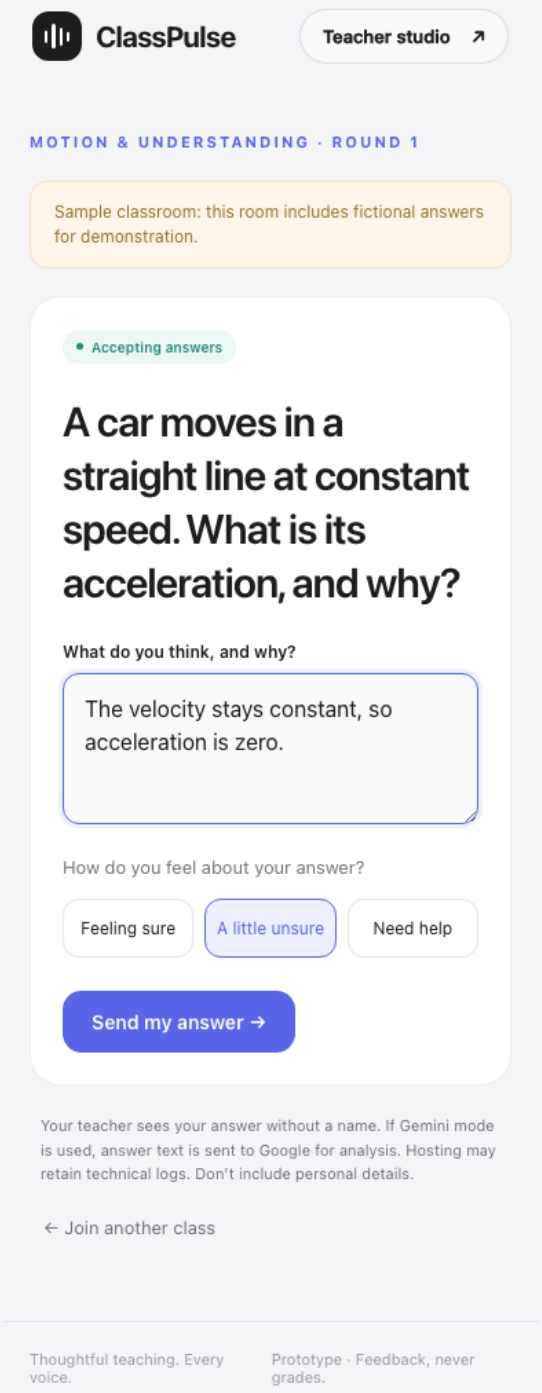
1. In Teacher studio, click Explore a sample class. Eight fictional answers appear, clearly labelled.

2. Copy the student link. Open a private/incognito window on the same laptop and paste it.

3. Type: The velocity stays constant, so acceleration is zero. Choose a confidence level and press Send my answer.

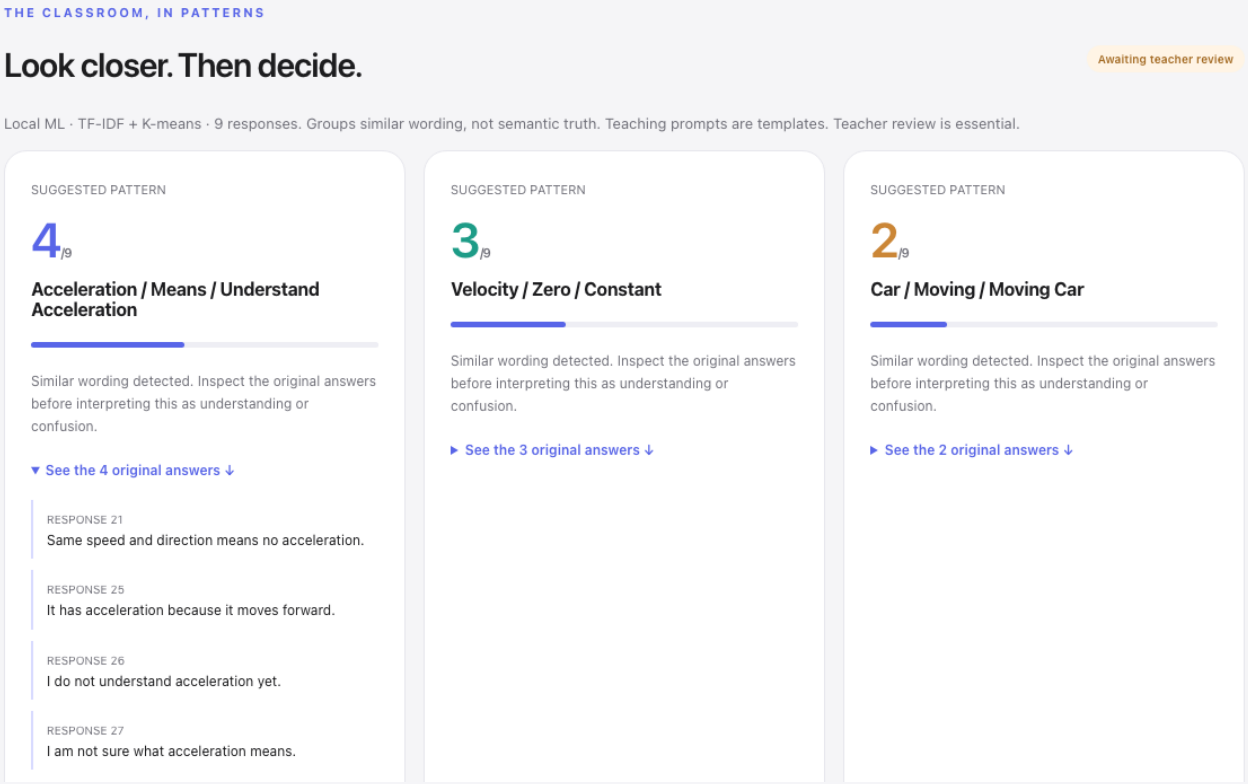
4. Return to the teacher window. Within a few seconds, the total changes from eight to nine. This extra answer came from your separate student session.

For a real class: create a new empty room instead of using the sample. Phones need a reachable hosted address; localhost on a phone means that phone.



Reveal. Inspect. Decide.

The original answers matter more than an impressive-looking chart.



1. Click Close this round, then Reveal patterns. Local ML works without an API key.

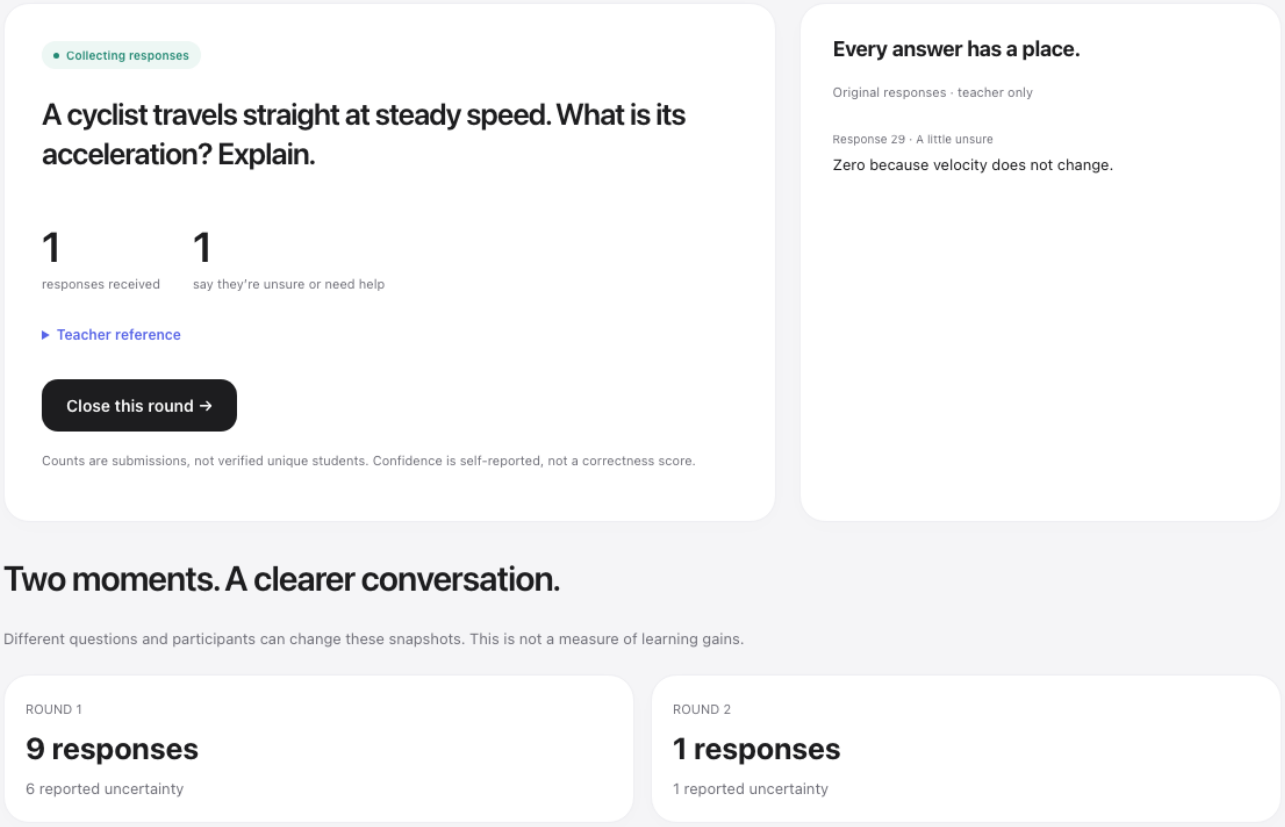
2. Expand the original answers under a group. Check every interpretation against the actual words. Similar wording can hide opposite meanings.

3. Write your interpretation or correction. Click Mark as reviewed, or reject the interpretation if it is wrong.

Explain honestly: local TF-IDF and K-means group similar words. Teaching prompts are templates. Optional Gemini mode needs a private server API key and has not been verified here with a live key.

Ask again. Close the loop.

Show how the teacher acts on the evidence, not just how a dashboard looks.



1. Click Edit & open the next round. Replace the suggested question with: A cyclist travels straight at steady speed. What is its acceleration? Explain.

2. Add the teacher reference: Zero, because velocity is constant. Open the round.

3. The student window receives the new question automatically. Send another answer. Check the live count and round comparison.

4. Export the room JSON if needed. The comparison shows submissions and self-reported uncertainty, not measured learning gains.

Present in four minutes.

Rehearse once. Keep the story small, concrete and honest.

0:00-0:30 - Problem. A raised hand does not reveal what the whole class understands. Quiet students need a low-pressure way to explain their thinking.

0:30-1:20 - Live answer. Show the labelled sample, then add one fresh answer in a separate student window. Point to the changing count.

1:20-2:30 - AI with evidence. Close, analyse and open the source answers. Explain the actual mode and one limitation. Show a teacher correction.

2:30-3:30 - Teacher action. Open a follow-up question and answer it. Explain why this is a feedback loop rather than an automatic grading tool.

3:30-4:00 - Finish. Name the stack: Python, Flask, SQLite and scikit-learn. State what is tested and what remains a prototype. Do not claim classroom results you have not measured.

Before submission: public app URL; public GitHub source and README; 3-5 minute video on Drive; all links work in incognito. GITHUB_SETUP.txt explains the browser-only upload steps.

Timing: the reported 11 am Pakistan start plus five hours gives 4 pm Pakistan on 23 October 2026. Follow official changes. No code pushes after the deadline.

Package status: a working local app and source files do not automatically create a public website, repository or uploaded video. Check docs/SUBMISSION_CHECKLIST.txt.